Windows Networking, Firewalls, Remote Services, and System Enumeration

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts is 3 hours.

Windows Networking Internals

I already know about ports, TCP/UDP, packets, ARP, DHCP, NAT, netstat, nmap

Run **ipconfig /all** but investigate every field.

Windows IP Configuration: the network settings for this Windows machine

Host Name: DESKTOP-4394834 is the computer's name on the network

Node Type: Hybrid. It affects local name resolution behavior.

IP Routing Enabled: No, This means your PC is NOT acting like a router. If it was yes, the computer could forward packets between networks and behave like a gateway/router

WINS Proxy Enabled: No. WINS is an OLD Windows name resolution system. Today is mostly legacy and rarely important in home environments.

Ethernet adapter Ethernet: These settings belong to the Ethernet (wired network card).

1. Media State: Media disconnected. This means the Ethernet adapter exists, but nothing is physically connected.
2. Description: Hardware/driver name of your network card
3. Physical Address (MAC address)
4. DHCP Enabled: Yes
5. Autoconfiguration Enabled: Yes, allows Windows to self-assign an address if DHCP fails

Wireless LAN adapter Wi-Fi: these settings belong to wireless network adapter

1. Connection-specific DNS Suffix: router/network automatically appends this for name resolution
2. Description: Wi-Fi hardware
3. Temporary IPv6 address: windows generates temporary IPv6 addresses for privacy because normal IPv6 can reveal device identity
4. Link-local IPV6 address: LOCAL only, cannot be routed on internet

There's more, but these are the few I needed to freshen up on.

Now, in Powershell run **nslookup google.com**, **nslookup discord.com,** and **nslookup github.com.**

What is recursive DNS?
The process where a DNS server does the work of finding a website's IP address FOR YOU. In recursive DNS, your computer asks ONE DNS server: "What is google.com?" That DNS server: contacts the other DNS servers, searches the internet hierarchy, gets the answer, and returns the final result to you.

What is caching in DNS?
A DNS server or computer temporarily REMEMBERS a domain name lookup so it does not have to search for it again.

What happens if DNS fails?
If DNS fails, your computer can still connect to the internet, but it cannot find websites by name. So you could reach google by typing the actual 32-bit IP address, but you cannot translate names into numbers.

Authoritative vs Non-authoritative:
Authoritative Answer is when the DNS server is the OFFICIAL source for that domain.
Non-authoritative Answer is when the DNS server got the answer from cache or another DNS server.

Run **route print.** This command displays your computer's routing table.

What is a routing table?
A list of rules your computer uses to decide: "Where should I send this network traffic?"
Every time your computer sends data: opening a website, playing VALORANT, etc….Windows checks the routing table first.

What is the default route?
The route your computer uses when it does NOT know exactly where to send traffic.

What does 0.0.0.0 mean in routing?
Any destination or unknown network. It will catch all bits since 0.0.0.0/0 represents everything.

Windows Firewall + Remote Services

Open it with: **wf.msc**

What is a firewall?
A security system that monitors and controls network traffic entering or leaving a device or network. It decides what traffic is allowed and what is blocked based on rules.

What is inbound vs outbound filtering?
Inbound = traffic coming INTO your computer/network
Outbound = traffic LEAVING your computer/network

What is a stateful inspection?
A firewall technique where the firewall remembers and tracks active network connections.

What is Remote Desktop Protocol (RDP)?
A Microsoft protocol that allows you to remotely control another computer over a network. It uses port TCP 3389.

What is SMB?
SMB stands for Server Message Block. Network protocol used mainly by Windows systems for: file sharing, printer sharing, shared folders, and network communication between computers. Computer A shares folder C:\Shared, SMB allows Computer B to access it over network \\DESKTOP-PC\Shared. SMB uses port TCP 445.

Run: **netstat -ano | findstr LISTENING.** Reminder: netstat command displays active network connections and networking information on your computer.

Why is exposing SMB dangerous?
If attackers can reach SMB: they may steal files, guess passwords, move through networks, execute malware, and fully compromise systems.

Run: **Get-NetFirewallProfile**

Deep System Enumeration

Enumeration = systematically gathering information about a system.

Run: **systeminfo.**

Run: **whoami, whoami /priv, net user**

What are Windows privileges?
Special permissions that determine what actions a user, process, or program is allowed to perform on a Windows system.

What is least privilege?
Users, programs, and systems should only have the minimum permissions and privileges necessary to perform their job.

What is UAC?
User Account Control. A Windows security feature that helps prevent unauthorized or dangerous system changes by requiring approval before giving programs elevated privileges. It is the popup that says: “Do you want to allow this app to make changes to your device?”

Run: **net share**

What are administrative shares?
Hidden Windows network shares automatically created by Windows for system administration purposes. They allow administrators to remotely access drives, system folders, and management resources over the network using SMB.

Run: **schtasks**

Why do attackers abuse scheduled tasks?
They allow malware or malicious commands to run automatically, repeatedly, silently, or even after reboot. Persistence means: staying on the system after initial compromise.